

AI Event Coordinator: Intelligent System for Automated Event Planning and Schedule Management

Assignee : Samidha

Assigned by : Tharuna Sree

Assigned on : 08-05-2026

Project Description:

This project focuses on building a Generative AI-powered system that assists users in planning and organizing events efficiently through automated coordination, scheduling, and recommendation generation. The system takes user requirements such as event type, budget, location, audience size, and preferences, then generates a complete event plan including venue suggestions, timelines, activity schedules, and resource recommendations.

The AI Event Coordinator simplifies the traditionally time-consuming process of event management by transforming high-level user ideas into structured, actionable event plans. Instead of manually searching for venues, managing schedules, and coordinating logistics, users receive AI-generated event blueprints tailored to their requirements.

The system is designed as a modular planning engine where venue discovery, scheduling, budgeting, and itinerary generation are handled by independent AI-assisted tools connected through a central orchestration workflow.

Project Scenario:

Planning events such as workshops, conferences, birthdays, corporate meetings, or college festivals often requires managing multiple components simultaneously—venue selection, scheduling, guest coordination, budgeting, and activity planning. This process can become overwhelming, especially when users lack experience or time.

The AI Event Coordinator addresses this challenge by acting as a virtual planning assistant. When a user enters event details such as “Plan a one-day AI workshop for 100 students within a moderate budget,” the system generates:

- Suitable venue suggestions based on capacity and location
- Event schedules with sessions, breaks, and activities
- Budget estimations and resource planning
- Checklists for logistics and coordination

For example:

- “Suggested venue: Community Hall with projector facilities”
- “Recommended schedule: Registration → Keynote → Workshop → Networking Session”
- “Estimated catering and equipment requirements based on attendee count”

This enables users to organize events more efficiently while reducing planning complexity and coordination effort.

Key Functions and Modular Tools:

- ◆ **Requirement Collection & Context Understanding Tool**
→ Captures event details such as type, budget, attendee count, location, duration, and preferences.
- ◆ **Venue Recommendation Tool**
→ Uses search APIs and AI filtering to suggest venues based on capacity, facilities, accessibility, and budget constraints.
- ◆ **Schedule & Agenda Generator Tool**
→ Creates structured event timelines including sessions, breaks, activities, and task sequencing.
- ◆ **Budget Planning Tool**
→ Estimates costs for venue, catering, equipment, decorations, and logistics based on event scale.
- ◆ **Resource & Logistics Management Tool**
→ Suggests required resources such as seating, audio systems, projectors, staff, and transportation.
- ◆ **Checklist & Task Automation Tool**
→ Generates event preparation checklists and coordination tasks for smooth execution.
- ◆ **Structured Output Generation Tool**
→ Converts all generated plans into organized formats such as tables, schedules, JSON outputs, or printable reports.
- ◆ **Personalized Recommendation Tool**
→ Adapts plans based on event type (formal, casual, educational, corporate) and user preferences.

Tools and Technologies Used by the System:

-  **Python** – Core backend development and workflow orchestration
 -  **OpenAI GPT-4 / GPT-5 or Open-Source LLMs** – Event planning, scheduling, and recommendation generation
 -  **Search APIs** – Venue discovery, location suggestions, and event-related information retrieval
 -  **Prompt Engineering Techniques** – Structured planning prompts and contextual recommendations
 -  **Structured Output Generation** – JSON/table-based event plans and schedules
 -  **LangChain / Custom Pipelines** – Managing multi-step event planning workflows
 -  **Streamlit / React (Optional)** – Interactive interface for event customization and plan visualization
 -  **SQLite / Firebase (Optional)** – Storage for event plans, schedules, and user preferences

Files Need to Be Submitted:

- 📁 **Python Folder** (Modules for planning, venue recommendation, scheduling, and budgeting logic)
- 📁 **Frontend Folder (Optional)** (Interactive UI for creating and managing event plans)
- 📄 **Documentation** (System architecture, planning workflow, recommendation strategy, and structured output design)
- 🎥 **Video Demonstration** (End-to-end walkthrough showing event input → venue suggestions → schedule generation → final event plan)